```
        for x in range(2, 11):
            Musician.objects.create(
                first_name=f"First{x}",
                last_name=f"Last{x}",
                birth=date(1900, 1, x),
            )
```

With the new musician data in place, start by fetching all of them on a single page. The view defaults to 10 per page, so no query parameters should result in no pagination. The following listing validates this scenario.

Listing E.36 No query parameters and no pagination

```
# RiffMates/bands/tests.py
...
class TestBands(TestCase):
    ...
    def test_musicians(self):
        ...
        # Test URL with no arguments, 10 musicians means no pagination
        url = "/bands/musicians/"
        response = self.client.get(url)

        self.assertEqual(200, response.status_code)
        self.assertEqual(10,
            len(response.context["musicians"]))
        self.assertFalse(
            response.context["page"].has_previous())
        self.assertFalse(
            response.context["page"].has_next())
```

No query parameters in the URL

Always verify that 200 ("ok") came back.

Check there are 10 musicians in the context.

The .has_previous() call should return False.

The .has_next() call should return False.

Next, it is time to test using a query parameter to change the number of items on the page and that pagination works. The following listing sets the items per page to five, resulting in two pages.

Listing E.37 Test five musicians per page

```
# RiffMates/bands/tests.py
...
class TestBands(TestCase):
    ...
    def test_musicians(self):
        ...
        # Test with 5 per page
        url = "/bands/musicians/?items_per_page=5"
        response = self.client.get(url)

        self.assertEqual(200, response.status_code)
        self.assertEqual(5,
            len(response.context["musicians"]))
        self.assertFalse(response.context["page"].has_previous())
        self.assertTrue(response.context["page"].has_next())
```

Use the items_per_page query parameter to set five musicians per page.

Validate that the right number of musicians came back and that there is no previous page but there is a next page.